

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		(Primary structure is) the {sequence/ order} of amino acids (1)	1			1		
		(ii)		Secondary structure – alpha helices / beta pleated sheet (1) Tertiary structure – (secondary structure) {folded into/ held in} a {globular /3D} {structure/ shape} (1)		2		2		
		(iii)		{Bacteria absorb <u>water</u>/ <u>water</u> enters bacterial cells} by <u>osmosis</u> (1) {less negative/ higher} {water / solute} potential outside (than inside)/ lower solute concentration outside (than inside)/ reference to water potential gradient into cell(1) Accept ORA - Bacterial cells burst (due to damage to cell wall) (1)		3		3		
		(iv)		<u>substrate</u> changes the shape of the <u>active site</u> (as it enters it) / (enzyme/ lysozyme) changes shape to allow <u>substrate</u> to fit into <u>active site</u> / OWTTE (1)	1			1		
	(b)	(i)		Any three (×1) from A. Decreases away from optimum/ description of shape of full graph/ owtte (1) B. Of pH 5.2 (1) C. (changing pH) changes the {secondary/tertiary} structure/ (changing pH) {changes/ breaks} the bonds that hold the enzyme in shape/ owtte (1) Ignore reference to denaturing D. Disruption of active site prevents substrate binding/ fewer enzyme substrate complexes/ active site is no longer complementary to the substrate (1)	2	1		3	1	